



Reproducing the EEG-fMRI Pipeline in Python

EEGLAB reproduction, machine learning & MNE validation

R&D Internship — MIMLAB, Boğaziçi University

Aymen Khatrani

Supervisor: Prof. Pınar S. Özbay

ISIA tutor: C. Dhaenens

August 26, 2026



Outline

- ▶ Introduction
- ▶ Preprocessing pipeline
- ▶ Machine learning
- ▶ Downstream & conclusion



Plan

1 Introduction

- ▶ Introduction
- ▶ Preprocessing pipeline
- ▶ Machine learning
- ▶ Downstream & conclusion



Context & problem statement

Recording EEG inside the MRI scanner

Simultaneous EEG–fMRI combines two complementary modalities:

- **EEG** — millisecond temporal precision
- **fMRI** — millimetre spatial precision

But inside the scanner the EEG is buried under two artifacts.

Cleaning is classically done in **EEGLAB** (MATLAB, FMRIB plugin) — proprietary, manual, poorly reproducible.

The price to pay

Gradient artifact ≈ 1 mV ($\times 1000$ the useful EEG).

Pulse / BCG artifact driven by the heartbeat.

Can we redo it in Python: open, reproducible, and extendable?



Goals & objectives

1 Introduction

O1 — Reproduce

Recode each EEGLAB step in Python/MNE, including FASTR (gradient) and OBS (BCG), absent natively from MNE.

O2 — Build an ML dataset

Extract features describing the ICA components and label them.

ISIA competencies: advanced scientific knowledge · building learning datasets · model selection & optimization.

O3 — Model

Select and optimize a component classifier, benchmarked against ICLabel.

O4 — Validate

Quantitatively compare EEGLAB and Python (spectrum, wavelets, ICA, statistics).



Data — subject sub-38

Resting state, recorded inside the scanner

- **32 channels** (31 EEG in the 10–20 system + 1 ECG)
- Sampling rate **5000 Hz**
- Duration $\approx$ **392 s** per recording
- **2 runs** (run-O1, run-O2)

The main walk-through uses sub-38; the **bonus ML robustness studies** (method comparison, LOSO) pool 4–5 subjects (sub-37/38/41/55/56) for statistical credibility.

Also runs without data

A realistic **simulated** dataset (1/f + alpha EEG, ECG, injected gradient artifact) makes the notebook executable end-to-end for demos and tests.



Plan

2 Preprocessing pipeline

- ▶ Introduction
- ▶ **Preprocessing pipeline**
- ▶ Machine learning
- ▶ Downstream & conclusion



Preprocessing

2 Preprocessing pipeline

- Faithful chaining of the EEGLAB steps
- FASTR (gradient) and OBS (BCG) coded from scratch



The processing chain, step by step

2 Preprocessing pipeline

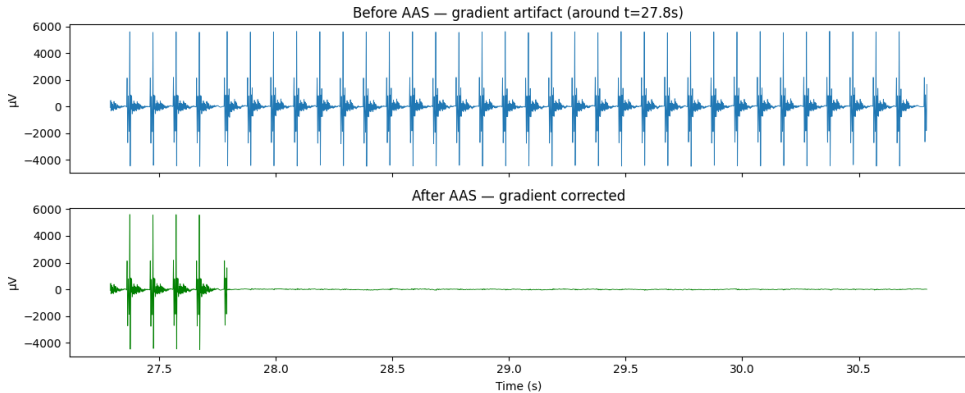
1. Trigger detection (data-driven)
2. Gradient artifact — **AAS**
3. 50 Hz notch filter
4. QRS detection on the ECG
5. Pulse artifact — **OBS**
6. Resampling to 250 Hz
7. Average reference (ECG excluded)
8. Band-pass 1–40 Hz
9. ICA — extended Infomax
10. ICLabel → removal → reconstruction

Trigger auto-detected: **R128**, $TR \approx 3$ s, 120 volumes.



Gradient artifact — AAS (Allen 2000)

Repetitive, volume-locked artifact averaged over a sliding window, then subtracted

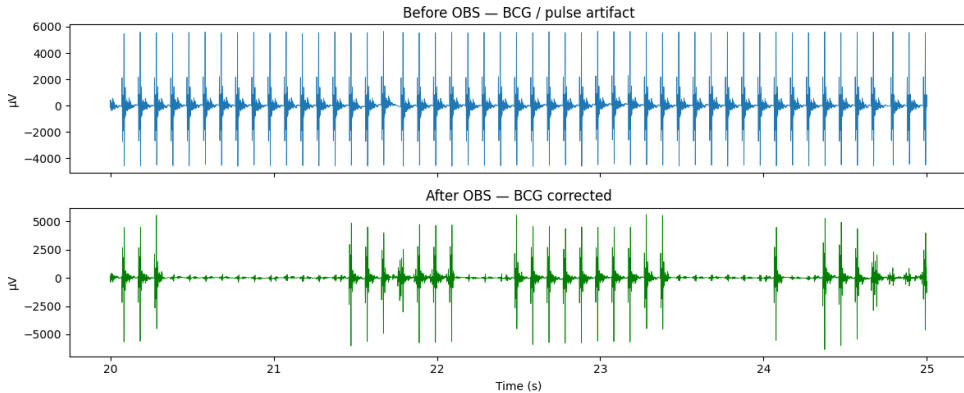


Before: ≈ 1 mV bursts phase-locked to the volume trigger. After: back to the μV range.



Pulse artifact — OBS (Niazy 2005)

Optimal Basis Set = mean artifact + 4 principal components (SVD), per channel



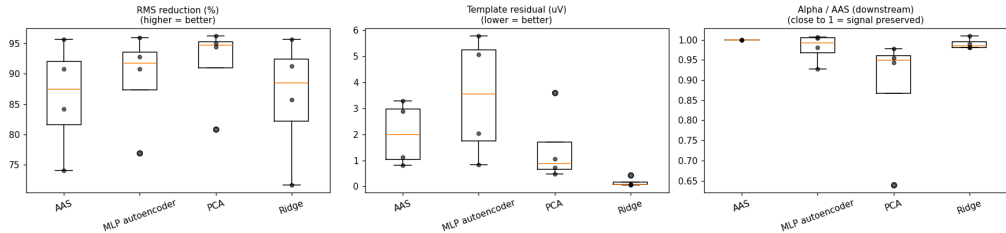
767 heartbeats detected on the ECG; the beat-varying BCG is projected onto the basis and removed.



Bonus — gradient: does ML beat AAS? (4 subjects, out-of-sample)

K-fold CV over occurrences: each method fitted on train folds, scored only on held-out occurrences

Method comparison (gradient) - 4 subjects, out-of-sample

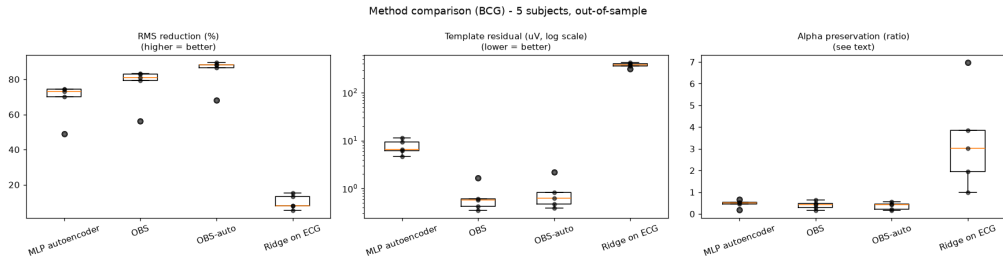


RMS reduction (%), absolute template residual (μV), and downstream alpha relative to AAS — 4 subjects, per-subject dots overlaid on the boxplots.



Bonus — BCG: does ML beat OBS? (5 subjects, out-of-sample)

Same out-of-sample protocol as the gradient, over heartbeats instead of volume occurrences



Note the log scale on the residual panel: OBS/OBS-auto sit near 0.3–2 μV , the naive Ridge-on-ECG near 300–400 μV .



Bonus — what the comparison teaches (corrected, multi-subject)

Supersedes the earlier single-subject, in-sample read of this bonus

Gradient: Ridge is the safe upgrade

Ridge wins the residual in 4/4 subjects (30–60× smaller than AAS) with alpha untouched (0.98–1.01). **PCA** wins RMS in 4/4 but risks alpha at sub-55 (0.64, –36%) — a real, subject-specific over-correction.

BCG: OBS still wins

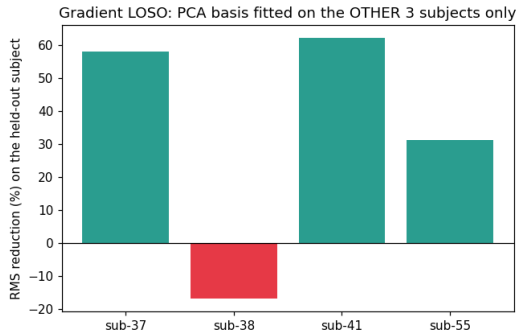
OBS-auto beats OBS on RMS in 5/5 but costs alpha in 5/5 without exception — overfitting, not real gain. The **MLP**, once scored out-of-sample, is the *least* reliable method (worst residual in 5/5) — reverses the earlier (leaky) result.

Wilcoxon at this N cannot cross $p = 0.05$ by construction (min. $p = 0.125$ at $N=4$, 0.0625 at $N=5$) — the per-subject win count above is the primary evidence.

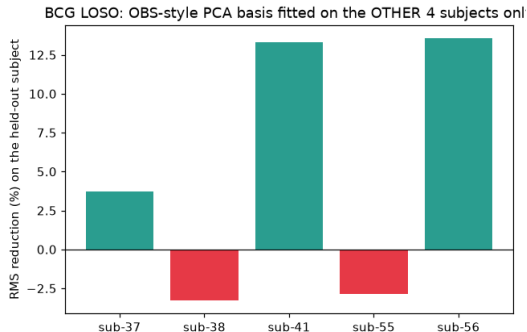


Bonus — does it generalize across subjects? (LOSO)

Basis fitted on the OTHER subjects only, applied unmodified to the held-out one



Gradient PCA basis: median 44.7%, but **sub-38 negative**



BCG OBS-style basis: 2/5 subjects negative

Compare to 79–96% within-subject reduction (1.3bis/1.6bis): transferring a basis across subjects destroys most of its value, and for some subjects makes the signal worse than doing nothing. Artifact correction is a per-recording calibration problem, not a learnable universal pattern.



Bonus — fixing the delay assumption & validating downstream

Two follow-ups on the BCG result above

Ridge-ECG needed a delay

The naive Ridge regressed on $ecg(t)$ (zero-delay) — physiologically wrong, the BCG lags the QRS by ~ 200 – 400 ms. Adding **61 time-shifted ECG copies** (-300 to $+600$ ms):

	naive	delayed
sub-37	15.2%	51.0%
sub-38	7.9%	47.0%
sub-56	8.0%	46.7%

5/5 subjects, 3 – $6\times$ better — but still short of OBS (79–90%).

Does RMS-winning help ICA?

Full chain to ICA + ICLabel, % components labelled brain (4 subjects):

Method	mean % brain
PCA	45.0
AAS	41.7
Ridge	41.7
MLP	36.7

PCA has the best mean but wins only 2/4 subjects head-to-head vs AAS — it loses *exactly* at sub-55 (alpha risk) and at sub-38, the main case-study subject.



Bonus — how close is our pipeline to the real EEGLAB?

EEGLAB's own FASTR and pop_fmrib_pas, exported mid-pipeline and scored with the same metric (4 subjects)

Gradient: FASTR is very good — and so is ours

Method	mean RMS red.
EEGLAB (FASTR)	93.0%
PCA	91.7%
MLP	89.1%
AAS	86.2%
Ridge	86.1%

FASTR has the best mean RMS reduction — but **Ridge still has the lowest residual at every subject** (0.05–0.44 μV vs EEGLAB's 0.8–2.9 μV).

BCG: the two metrics disagree

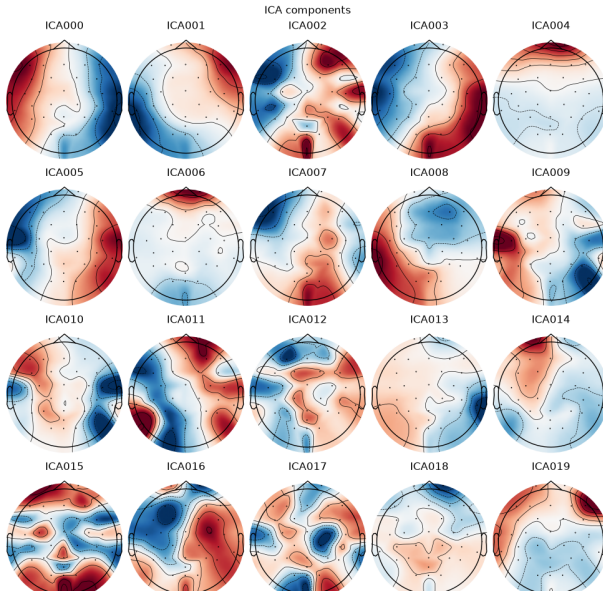
Method	RMS red.	residual (μV)
EEGLAB	92%	2.3–4.7
OBS (4 PC)	79–83%	0.4–1.7

EEGLAB (pop_fmrib_pas) wins RMS in 4/4 subjects, but the from-scratch **OBS has a 4–6 $\times$ lower residual in 4/4** — same RMS-vs-residual split as throughout this bonus, still in OBS's favour.



ICA & automatic classification (ICLabel)

2 Preprocessing pipeline

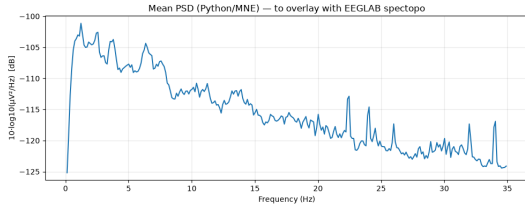


- Extended Infomax (= runica)
- Average reference → **30 components**
- ICLabel flags **5/30** as artifacts (eye blink, muscle)
- Conservative rejection (the other class is kept)

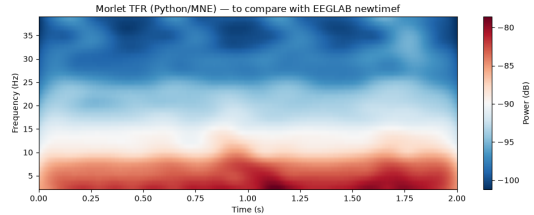


Spectral validation (Python side)

Cleaned signal follows the expected $1/f$ decay — ready to overlay with EEGLAB



PSD — Welch



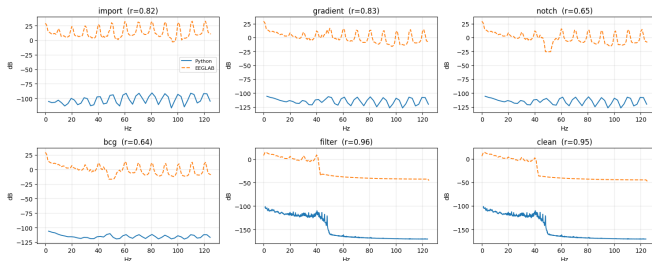
Time-frequency — Morlet



EEGLAB ↔ Python — quantitative equivalence (§2.5)

Driven by a real EEGLAB export — not a placeholder: computed, stage by stage

Stage-by-stage mean PSD — EEGLAB vs Python



- Per-stage PSD correlation rises to **0.96** (filter) / **0.95** (clean)
- Lowest at the correction stages (notch 0.65, BCG 0.64) — where the two implementations differ most
- Alpha peak **7.7 vs 7.1 Hz** — PASS
- ICA topographies median $|r| = 0.61$ (10 strong pairs)
- ICLabel agreement $\kappa = 0.58$ — PASS

The constant vertical offset is a unit convention (μV^2 vs V^2); the correlation compares spectral *shape*, which is scale-invariant.



Plan

3 Machine learning

- ▶ Introduction
- ▶ Preprocessing pipeline
- ▶ **Machine learning**
- ▶ Downstream & conclusion



Our own ICA-component classifier

We do not just call ICLabel — we build a dataset and train models

- **12 features / component:** spectral (band power, peak, 1/f slope), temporal (kurtosis, skewness), spatial (topography entropy).
- **Labels:** from ICLabel (cheap, but caps performance at “imitate ICLabel”).
- **Synthetic data:** 300 components generated for *training*; the 30 real ones reserved for *testing*.
- **4 model families:** logistic regression, random forest, SVM, gradient boosting — GridSearchCV, 5-fold CV.
- **Benchmark:** agreement with ICLabel via Cohen's κ .



Model comparison

Trained/selected on synthetic data only; the 30 real components are a pure out-of-sample test

Model	Train	CV	Synth test	Real OOS	Real OOS
	AUC	AUC	AUC	AUC	F1
LogReg	1.00	1.00	1.00	0.67	0.30
SVM	1.00	1.00	1.00	0.62	0.29
RandomForest	1.00	1.00	1.00	0.52	0.29
GradBoost	1.00	1.00	1.00	0.50	0.29

Perfect synthetic scores, but real out-of-sample AUC collapses towards chance.



Key result — an overfitting unmasked

3 Machine learning

Naive version (misleading)

Test AUC 0.999

κ vs ICLabel 0.71

Data leakage + trivial synthetic test set.

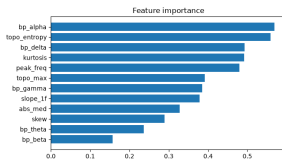
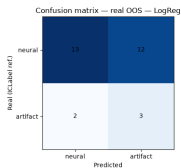
Corrected protocol (honest)

Synthetic AUC 1.00

Real AUC (OOS) 0.50–0.67

κ vs ICLabel 0.067

Trained on synthetic only; real never seen.



The model **overfit** the **artificial structure** of the synthetic data: it does not transfer to real components. A **negative but informative** result — a credible classifier needs real, multi-subject annotations.



Plan

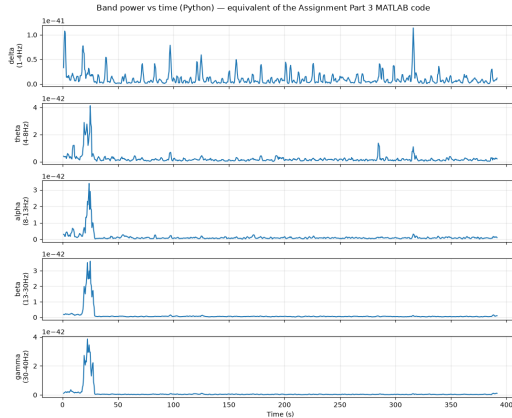
4 Downstream & conclusion

- ▶ Introduction
- ▶ Preprocessing pipeline
- ▶ Machine learning
- ▶ Downstream & conclusion



Downstream analysis — band power over time

Cleaned signal is directly usable; also a final quality check



Python reproduction of the Assignment Part 3 spectrogram. Marked transient spikes would betray a residual artifact.



Assumed limitations

4 Downstream & conclusion

- **Gradient / BCG:** simplified in-house AAS and OBS (not the full FASTR regularization).
- **Non-deterministic ICA:** structural comparison, not point by point.
- **Modest N (4-5 subjects):** enough to detect real effects and cross-subject failure (LOSO), but Wilcoxon at this N cannot cross $p = 0.05$ by construction — descriptive evidence (win count, median) is the priority.
- **Synthetic → real transfer** (component classifier): does not work; real multi-subject annotations are needed.
- **Spectral comparisons:** align windowing and units before any conclusion.



Conclusion & perspectives

4 Downstream & conclusion

Achieved

Complete EEG-fMRI pipeline reproduced in Python (FASTR & OBS from scratch)

EEGLAB $\leftrightarrow$ Python equivalence **quantified** (§2.5): PSD, ICA, ICLabel κ

Sound ML evaluation that unmasks over-optimism (component classifier *and* the artifact-correction bonus)

Multi-subject, out-of-sample comparison (4–5 subj.): Ridge/PCA credibly beat AAS, OBS still beats MLP

LOSO + delay-aware Ridge-ECG + ICLabel: correction is per-recording, not transferable

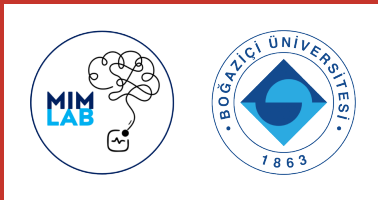
Perspectives

Manually annotate the real components (true ground truth vs ICLabel)

Extend LOSO / ICLabel-downstream validation to the BCG methods and to more subjects

Non-linear delay model for Ridge-ECG (still short of OBS with a linear FIR)

Make the synthetic generator more realistic



Thank you for your attention!

Aymen Khatrani — MIMLAB, Boğaziçi University